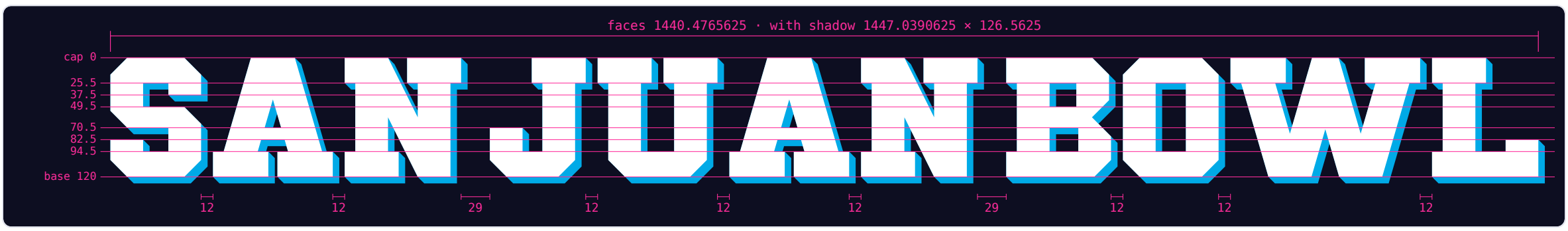
SAN JUAN BOWL LETTERING SPECIFICATION

The construction rules behind `SJB-Lettering-Exact.svg`. Every letter is built from the same small set of numbers, every coordinate is an exact decimal, and nothing is rounded. Values are in units of **u = cap height ÷ 120**. Pink lines are dimensions.

1 · The wordmark and its horizon lines



Eight shared horizontal lines. Every flat edge in every letter sits on one of them. Gaps under the wordmark are face to face: 12 between letters, 29 between words.

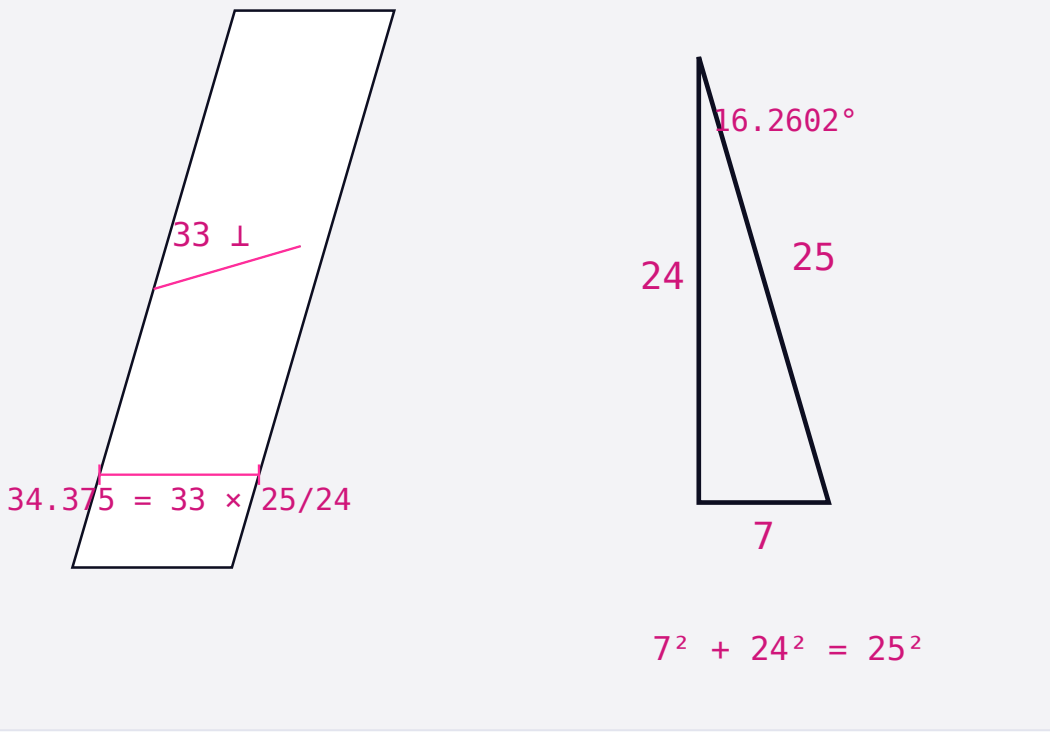
y (u)	Line	Used by
0	Cap line	Every letter top
25.5	Upper serif line	All top serifs and slabs; S top bar; B and O counter tops; U and J inner serifs
37.5	S upper terminal	S
49.5	Middle bar top	S, B (upper counter base)
70.5	Middle bar base	S, B (lower counter top); A counter base; J hook top
82.5	Crossbar line	A crossbar underside; L spur top; S lower terminal
94.5	Lower serif line	All foot serifs and slabs; B and O counter bases; U and J counter bases; L slab top
120	Baseline	Every letter bottom

2 · Units and proportions

Scale the wordmark by its cap height; every other measurement follows. The exact column is the value as a fraction, so it can be rebuilt at any size without drift.

Parameter	Value (u)	% of cap	Exact	Rule
Unit	`1 u = cap ÷ 120`	`0.8333 %`	—	Every value on this sheet is in u.
Cap height	`120`	`100 %`	`120`	Cap line y 0, baseline y 120 (y runs down, as in SVG).
Stem / stroke weight	`33`	`27.5 %`	`33`	Every vertical stem and every 7:24 stroke, measured perpendicular to the stroke.
Slab / bar depth	`25.5`	`21.25 %`	`51/2`	Top and bottom bars, serif slabs, B bowls, O and U bottoms.
Middle bar (S, B)	`21`	`17.5 %`	`21`	49.5 → 70.5
A crossbar	`12`	`10 %`	`12`	70.5 → 82.5
Serif overhang	`10.7265625`	`8.9388 %`	`1373/128`	= 31/24 × shadow + 2 (see 4).
Corner cut	`17 × 17`	`14.1667 %`	`17`	Every chamfer, exactly 45°. B waist notch is the one exception.
Lean (A, W)	`7 across : 24 down`	—	`7:24:25`	atan(7/24) = 16.260205°. Horizontal width of a 33 stroke = 34.375 exactly.
N diagonal	`30 across : 60 down`	—	`1:2`	Parallel band, perpendicular weight √720 = 26.832816.
Shadow vector	`(6.5625, 6.5625)`	`5.4688 % each axis`	`105/16`	Exactly 45°. Length 6.5625 × √2 = 9.280776.
Letter gap	`12`	`10 %`	`12`	Face to face, bounding box. Also the U opening.
Word space	`29`	`24.1667 %`	`29`	Face to face.
Overall, faces	`1440.4765625`	`1200.3971 %`	`184381/128`	S left edge to L right edge.
Overall, with shadow	`1447.0390625 × 126.5625`	—	—	Ratio 11.4334 : 1.

3 · Strokes and angles



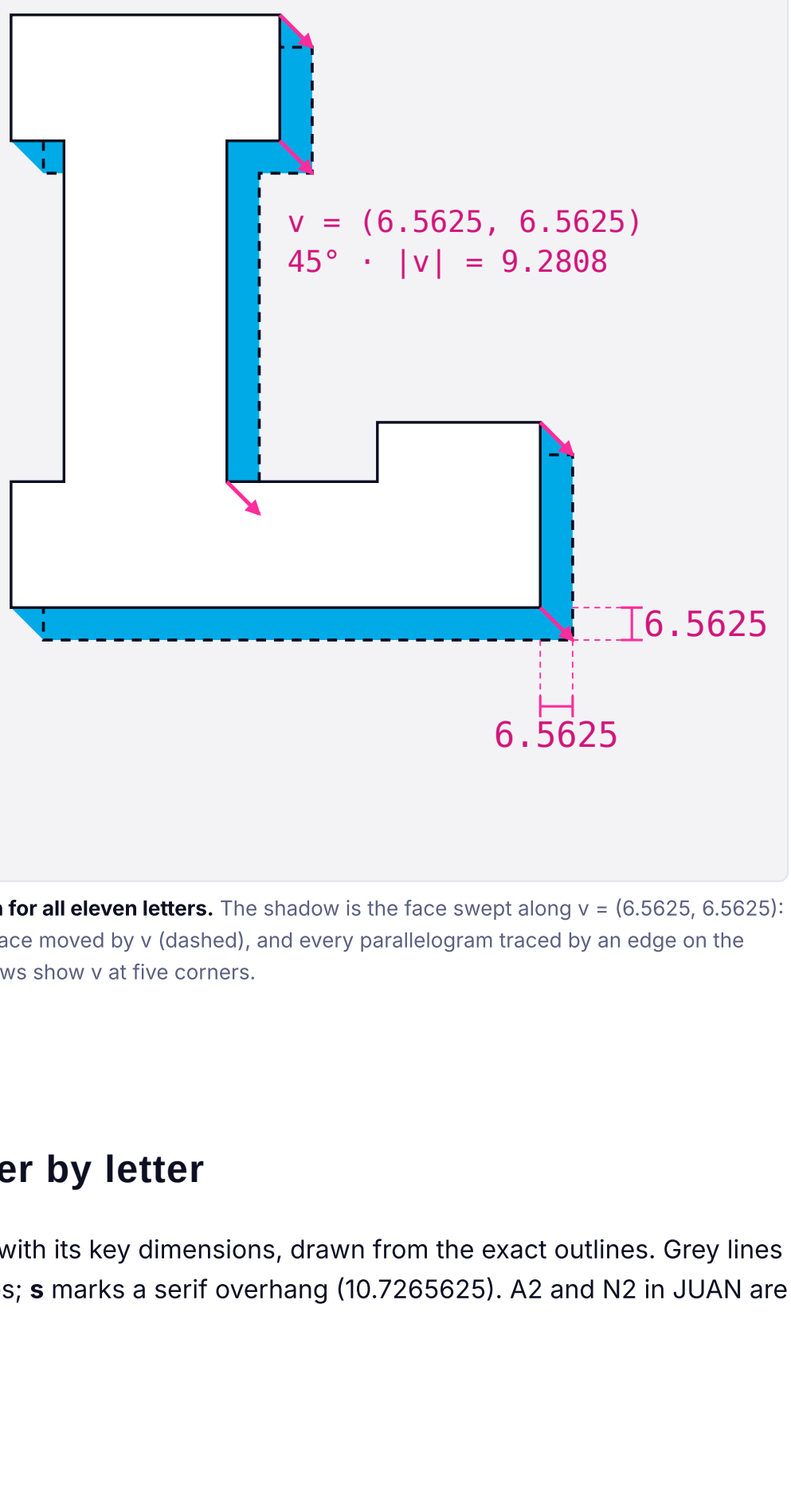
The 7:24 lean. A and W strokes lean 7 across for every 24 down, the 7-24-25 right triangle. Because 25/24 is exact, a 33 stroke is exactly 34.375 wide measured horizontally.

Only five edge directions exist in the whole wordmark.

Direction	Slope	Where	Count
Horizontal	0	bars, slabs, serifs	86
Vertical	∞	stems, serif ends	72
45°	±1	every corner cut, B waist	18
7:24	24/7 (73.7398° from horizontal)	A legs, W strokes	20
1:2	2 (63.4349° from horizontal)	N diagonal	4

```
stroke (perpendicular)   = 33
7:24 stroke, horizontal  = 33 × 25/24 = 34.375
N diagonal, perpendicular = 60/√5 = √720 = 26.832816
angle(7:24)  = atan(7/24) = 16.260205° from vertical
```

4 · Serifs, corners and the shadow



One extrusion for all eleven letters. The shadow is the face swept along v = (6.5625, 6.5625): the face, the face moved by v (dashed), and every parallelogram traced by an edge on the way. Pink arrows show v at five corners.

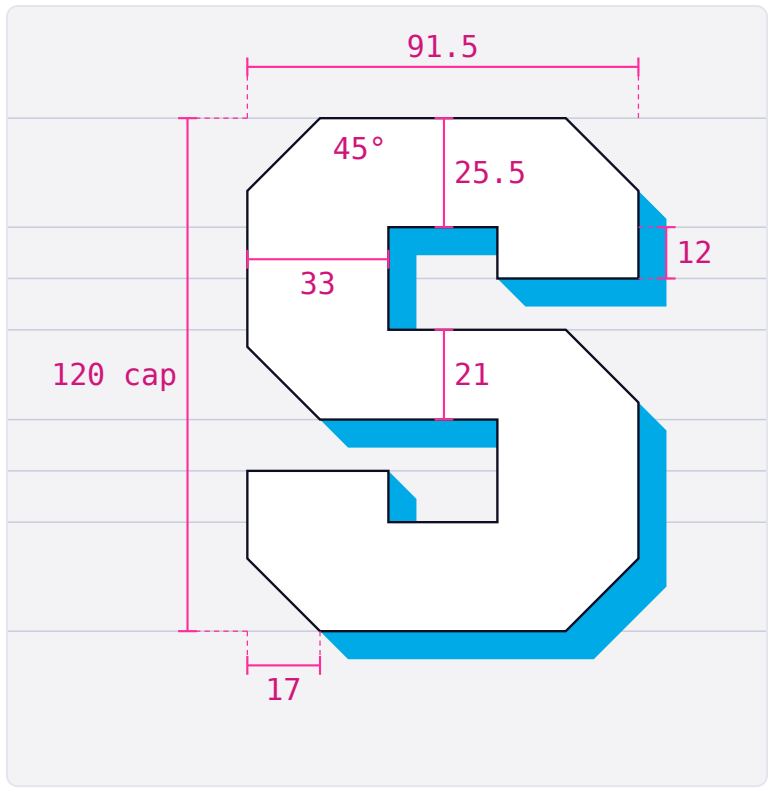
The shadow runs at exactly 45°, the same angle as every corner cut. So a lower-left cut casts no shadow at all instead of a hairline. The serif size is set by the shadow so that no shadow ever touches or nearly touches another shadow or a serif tip.

```
shadow v        = (6.5625, 6.5625)        |v| = 9.280776
serif s         = 31/24 × 6.5625 + 2 = 10.7265625
  31/24 × v     = 8.4765625, where a 7:24 leg
                  shadow reaches the foot
  + 2           = clearance kept at every serif
corner cut      = 17 × 17 at 45°
visible strip   = |v × d| / |d|   for an edge of direction d
```

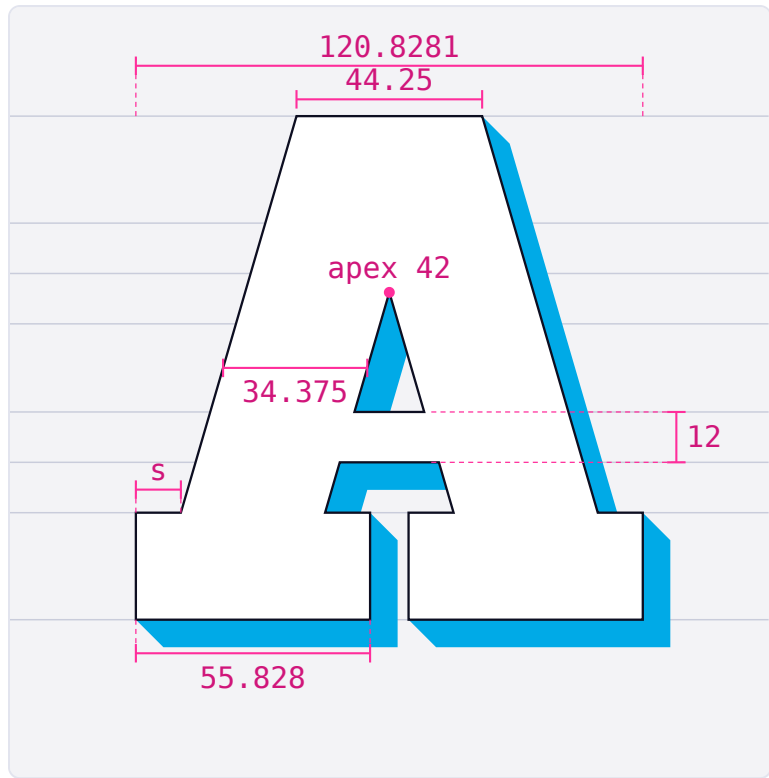
Edge	Direction d	Strip width	u
Horizontal (faces up/down)	(1, 0)	|v_y| = 6.5625	6.5625
Vertical (faces right)	(0, 1)	|v_x| = 6.5625	6.5625
45° cut at a lower-right corner	(1, −1)	(6.5625 + 6.5625)/√2	9.2808
45° cut at a lower-left or upper-right corner	(1, 1)	v ∥ edge, so 0	0 (no shadow)
7:24 edge slanting down-right, facing right	(7, 24)	6.5625 × (24 − 7)/25	4.4625
7:24 edge slanting down-left, facing right	(−7, 24)	6.5625 × (24 + 7)/25	8.1375
N diagonal	(1, 2)	6.5625 × (2 − 1)/√5	2.9348

5 · Letter by letter

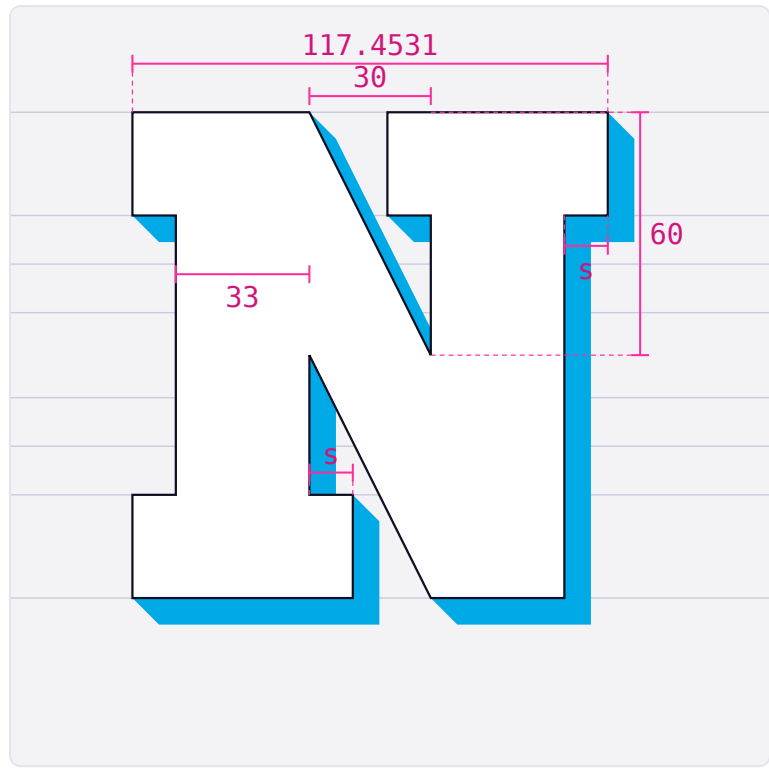
Each letter with its key dimensions, drawn from the exact outlines. Grey lines are the eight horizon lines; **s** marks a serif overhang (10.7265625). A2 and N2 in JUAN are exact copies of A and N.



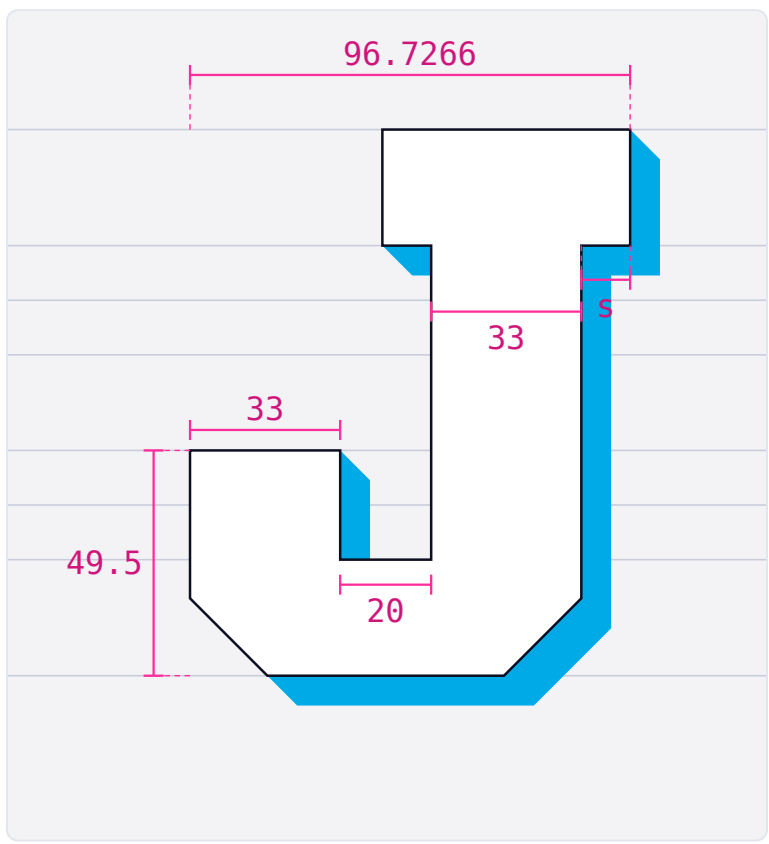
S · advance 91.5 · 22 anchors



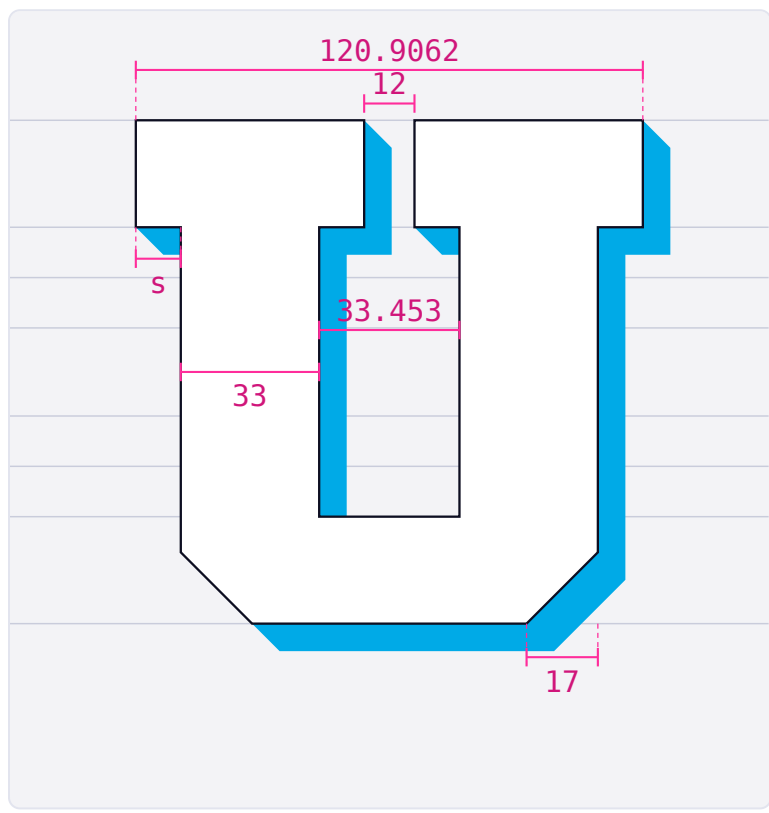
A (A2 identical) · advance 120.8281 · 19 anchors



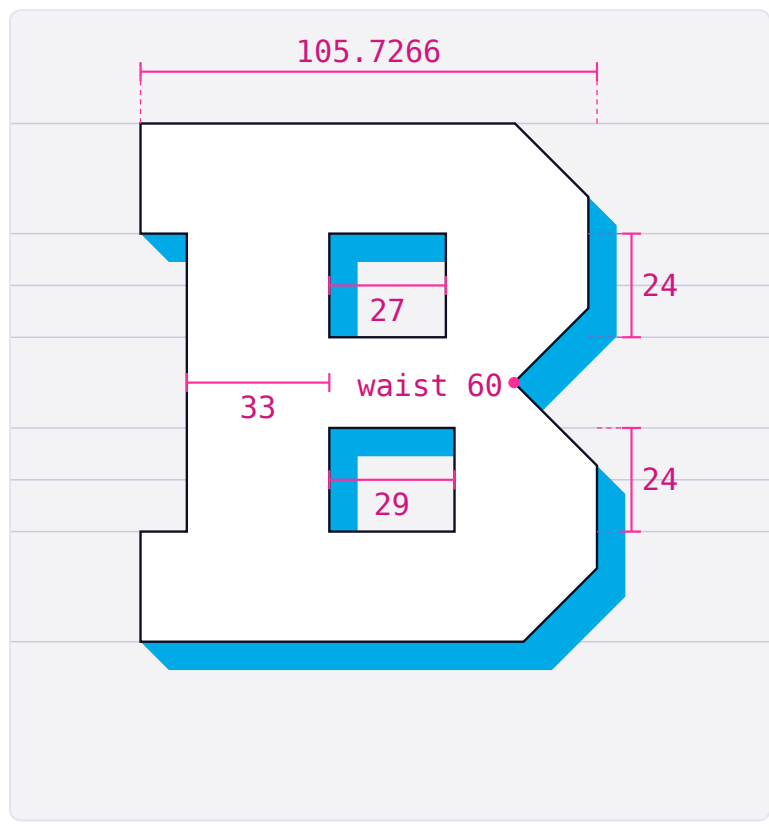
N (N2 identical) · advance 117.4531 · 20 anchors



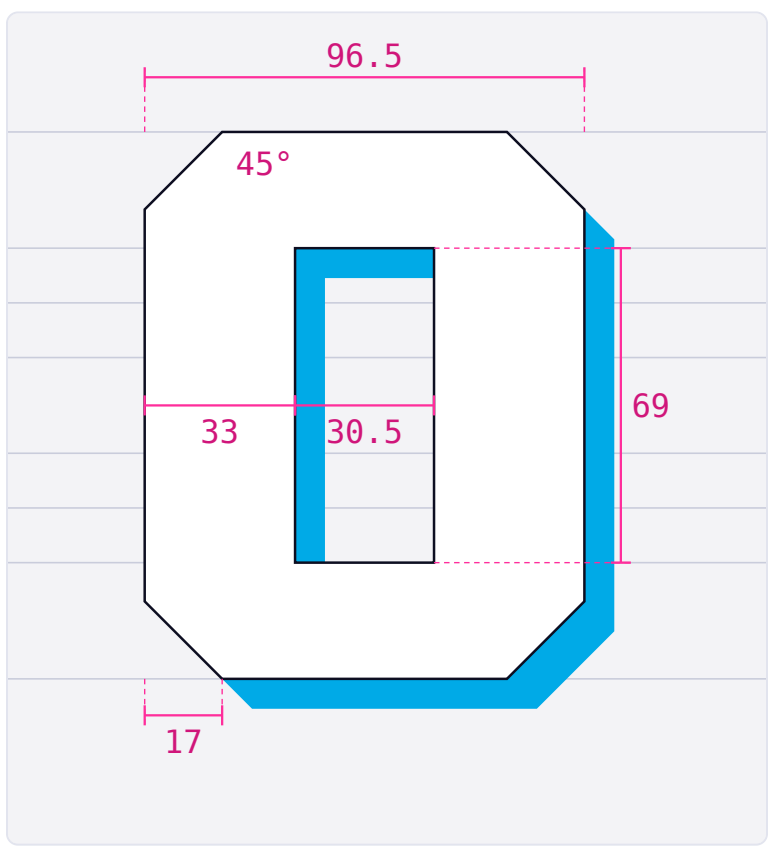
J · advance 96.7266 · 14 anchors



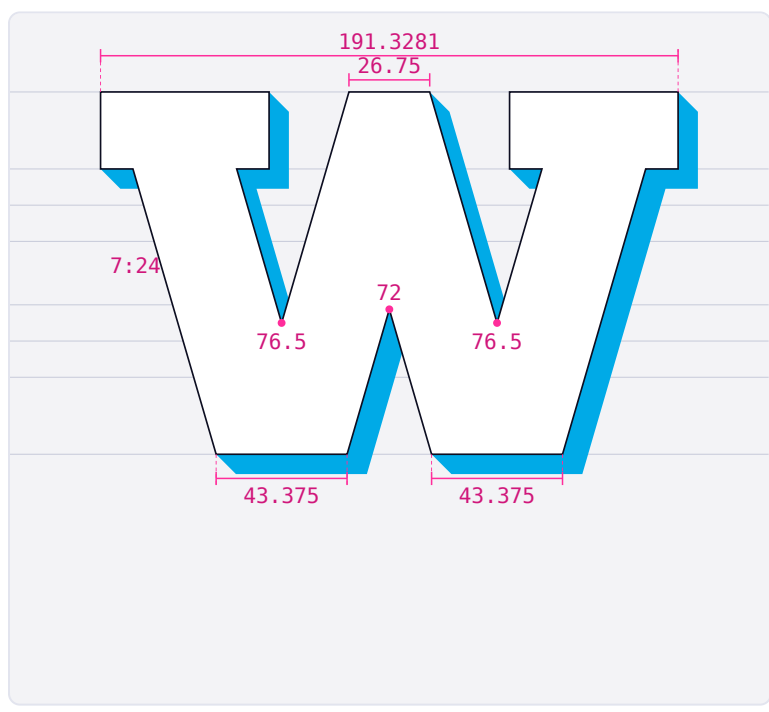
U · advance 120.9062 · 18 anchors



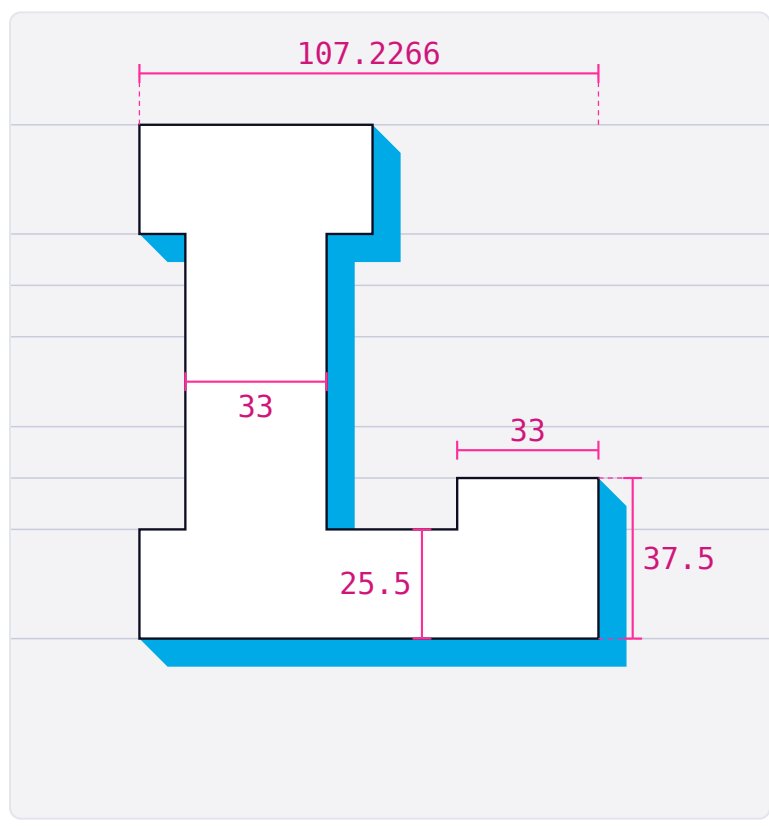
B · advance 105.7266 · 21 anchors



O · advance 96.5 · 12 anchors



W · advance 191.3281 · 21 anchors



L · advance 107.2266 · 14 anchors

Anchor coordinates

Every anchor of every letter in its own coordinates (left edge of the face at x 0, cap at y 0). Add the letter's x position from section 6 to place it.

S — 22 anchors, advance 91.5

Contour	#	x	y	Next edge	Length
outline	1	33	25.5	H	25.5000
outline	2	58.5	25.5	V	12.0000
outline	3	58.5	37.5	H	33.0000
outline	4	91.5	37.5	V	20.5000
outline	5	91.5	17	45°	24.0416
outline	6	74.5	0	H	57.5000
outline	7	17	0	45°	24.0416
outline	8	0	17	V	36.5000
outline	9	0	53.5	45°	24.0416
outline	10	17	70.5	H	41.5000
outline	11	58.5	70.5	V	24.0000
outline	12	58.5	94.5	H	25.5000
outline	13	33	94.5	V	12.0000
outline	14	33	82.5	H	33.0000
outline	15	0	82.5	V	20.5000
outline	16	0	103	45°	24.0416
outline	17	17	120	H	57.5000
outline	18	74.5	120	45°	24.0416
outline	19	91.5	103	V	36.5000
outline	20	91.5	66.5	45°	24.0416
outline	21	74.5	49.5	H	41.5000
outline	22	33	49.5	V	24.0000

A — 19 anchors, advance 120.828125

Contour	#	x	y	Next edge	Length
outline	1	0	94.5	H	10.7266
outline	2	10.7265625	94.5	7:24	98.4375
outline	3	38.2890625	0	H	44.2500
outline	4	82.5390625	0	7:24	98.4375
outline	5	110.1015625	94.5	H	10.7266
outline	6	120.828125	94.5	V	25.5000
outline	7	120.828125	120	H	55.8281
outline	8	65	120	V	25.5000
outline	9	65	94.5	H	10.7266
outline	10	75.7265625	94.5	7:24	12.5000
outline	11	72.2265625	82.5	H	23.6250
outline	12	48.6015625	82.5	7:24	12.5000
outline	13	45.1015625	94.5	H	10.7266
outline	14	55.828125	94.5	V	25.5000
outline	15	55.828125	120	H	55.8281
outline	16	0	120	V	25.5000
counter 1	1	52.1015625	70.5	7:24	29.6875
counter 1	2	60.4140625	42	7:24	29.6875
counter 1	3	68.7265625	70.5	H	16.6250

N — 20 anchors, advance 117.453125

Contour	#	x	y	Next edge	Length
outline	1	0	0	H	43.7266
outline	2	43.7265625	0	1:2	67.0820
outline	3	73.7265625	60	V	34.5000
outline	4	73.7265625	25.5	H	10.7266
outline	5	63	25.5	V	25.5000
outline	6	63	0	H	54.4531
outline	7	117.453125	0	V	25.5000
outline	8	117.453125	25.5	H	10.7266
outline	9	106.7265625	25.5	V	94.5000
outline	10	106.7265625	120	H	33.0000
outline	11	73.7265625	120	1:2	67.0820
outline	12	43.7265625	60	V	34.5000
outline	13	43.7265625	94.5	H	10.7266
outline	14	54.453125	94.5	V	25.5000
outline	15	54.453125	120	H	54.4531
outline	16	0	120	V	25.5000
outline	17	0	94.5	H	10.7266
outline	18	10.7265625	94.5	V	69.0000
outline	19	10.7265625	25.5	H	10.7266
outline	20	0	25.5	V	25.5000

J — 14 anchors, advance 96.7265625

Contour	#	x	y	Next edge	Length
outline	1	42.2734375	0	H	54.4531
outline	2	96.7265625	0	V	25.5000
outline	3	96.7265625	25.5	H	10.7266
outline	4	86	25.5	V	77.5000
outline	5	86	103	45°	24.0416
outline	6	69	120	H	52.0000
outline	7	17	120	45°	24.0416
outline	8	0	103	V	32.5000
outline	9	0	70.5	H	33.0000
outline	10	33	70.5	V	24.0000
outline	11	33	94.5	H	20.0000
outline	12	53	94.5	V	69.0000
outline	13	53	25.5	H	10.7266
outline	14	42.2734375	25.5	V	25.5000

U — 18 anchors, advance 120.90625

Contour	#	x	y	Next edge	Length
outline	1	0	0	H	54.4531
outline	2	54.453125	0	V	25.5000
outline	3	54.453125	25.5	H	10.7266
outline	4	43.7265625	25.5	V	69.0000
outline	5	43.7265625	94.5	H	33.4531
outline	6	77.1796875	94.5	V	69.0000
outline	7	77.1796875	25.5	H	10.7266
outline	8	66.453125	25.5	V	25.5000
outline	9	66.453125	0	H	54.4531
outline	10	120.90625	0	V	25.5000
outline	11	120.90625	25.5	H	10.7266
outline	12	110.1796875	25.5	V	77.5000
outline	13	110.1796875	103	45°	24.0416
outline	14	93.1796875	120	H	65.4531
outline	15	27.7265625	120	45°	24.0416
outline	16	10.7265625	103	V	77.5000
outline	17	10.7265625	25.5	H	10.7266
outline	18	0	25.5	V	25.5000

B — 21 anchors, advance 105.7265625

Contour	#	x	y	Next edge	Length
outline	1	0	0	H	86.7266
outline	2	86.7265625	0	45°	24.0416
outline	3	103.7265625	17	V	25.7734
outline	4	103.7265625	42.7734375	45°	24.3620
outline	5	86.5	60	45°	27.1905
outline	6	105.7265625	79.2265625	V	23.7734
outline	7	105.7265625	103	45°	24.0416
outline	8	88.7265625	120	H	88.7266
outline	9	0	120	V	25.5000
outline	10	0	94.5	H	10.7266
outline	11	10.7265625	94.5	V	69.0000
outline	12	10.7265625	25.5	H	10.7266
outline	13	0	25.5	V	25.5000
counter 1	1	43.7265625	25.5	H	27.0000
counter 1	2	70.7265625	25.5	V	24.0000
counter 1	3	70.7265625	49.5	H	27.0000
counter 1	4	43.7265625	49.5	V	24.0000
counter 2	1	43.7265625	70.5	H	29.0000
counter 2	2	72.7265625	70.5	V	24.0000
counter 2	3	72.7265625	94.5	H	29.0000
counter 2	4	43.7265625	94.5	V	24.0000

O — 12 anchors, advance 96.5

Contour	#	x	y	Next edge	Length
outline	1	17	0	H	62.5000
outline	2	79.5	0	45°	24.0416
outline	3	96.5	17	V	86.0000
outline	4	96.5	103	45°	24.0416
outline	5	79.5	120	H	62.5000
outline	6	17	120	45°	24.0416
outline	7	0	103	V	86.0000
outline	8	0	17	45°	24.0416
counter 1	1	33	25.5	H	30.5000
counter 1	2	63.5	25.5	V	69.0000
counter 1	3	63.5	94.5	H	30.5000
counter 1	4	33	94.5	V	69.0000

W — 21 anchors, advance 191.328125

Contour	#	x	y	Next edge	Length
outline	1	0	0	H	55.8281
outline	2	55.828125	0	V	25.5000
outline	3	55.828125	25.5	H	10.7266
outline	4	45.1015625	25.5	7:24	53.1250
outline	5	59.9765625	76.5	7:24	79.6875
outline	6	82.2890625	0	H	26.7500
outline	7	109.0390625	0	7:24	79.6875
outline	8	131.3515625	76.5	7:24	53.1250
outline	9	146.2265625	25.5	H	10.7266
outline	10	135.5	25.5	V	25.5000
outline	11	135.5	0	H	55.8281
outline	12	191.328125	0	V	25.5000
outline	13	191.328125	25.5	H	10.7266
outline	14	180.6015625	25.5	7:24	98.4375
outline	15	153.0390625	120	H	43.3750
outline	16	109.6640625	120	7:24	50.0000
outline	17	95.6640625	72	7:24	50.0000
outline	18	81.6640625	120	H	43.3750
outline	19	38.2890625	120	7:24	98.4375
outline	20	10.7265625	25.5	H	10.7266
outline	21	0	25.5	V	25.5000

L — 14 anchors, advance 107.2265625

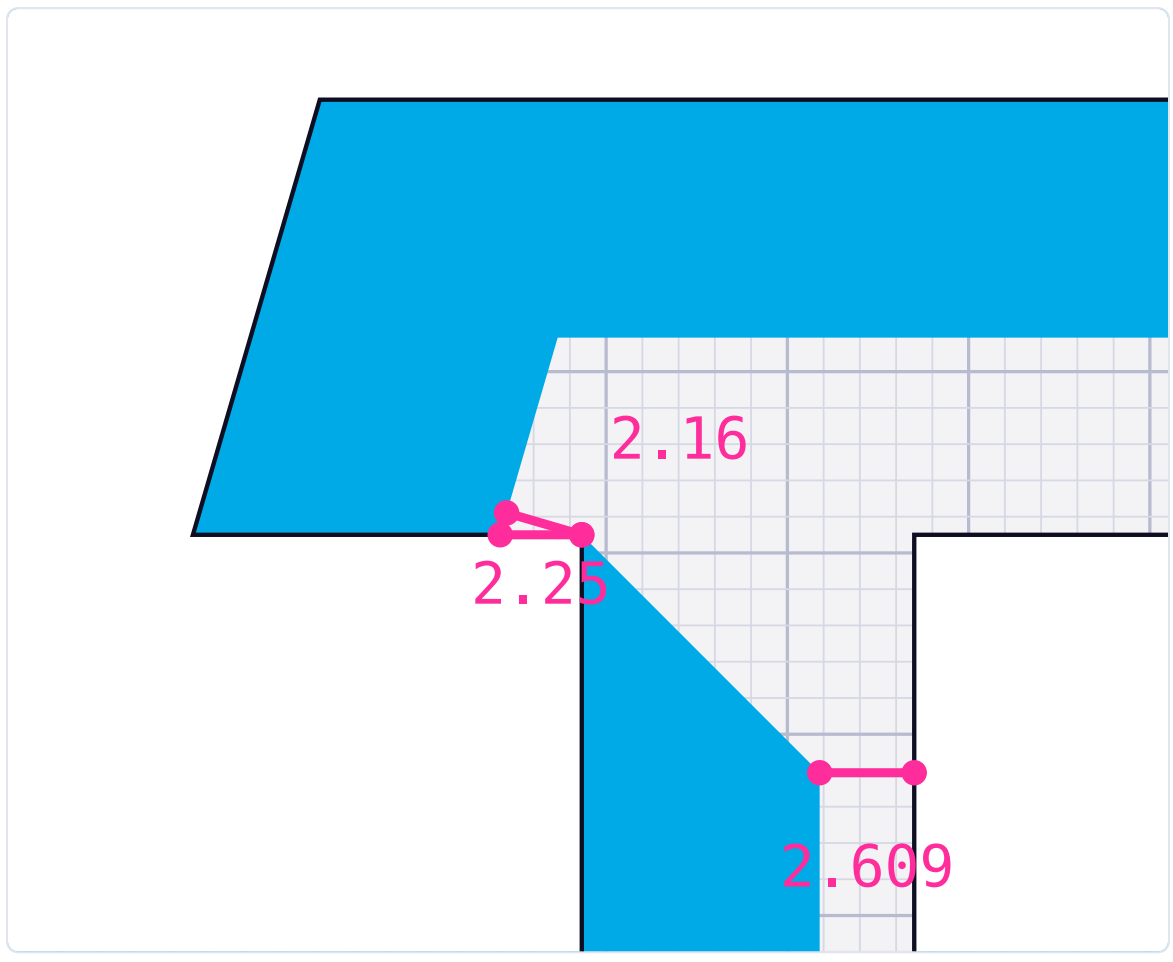
Contour	#	x	y	Next edge	Length
outline	1	0	0	H	54.4531
outline	2	54.453125	0	V	25.5000
outline	3	54.453125	25.5	H	10.7266
outline	4	43.7265625	25.5	V	69.0000
outline	5	43.7265625	94.5	H	30.5000
outline	6	74.2265625	94.5	V	12.0000
outline	7	74.2265625	82.5	H	33.0000
outline	8	107.2265625	82.5	V	37.5000
outline	9	107.2265625	120	H	107.2266
outline	10	0	120	V	25.5000
outline	11	0	94.5	H	10.7266
outline	12	10.7265625	94.5	V	69.0000
outline	13	10.7265625	25.5	H	10.7266
outline	14	0	25.5	V	25.5000

6 · Spacing and positions

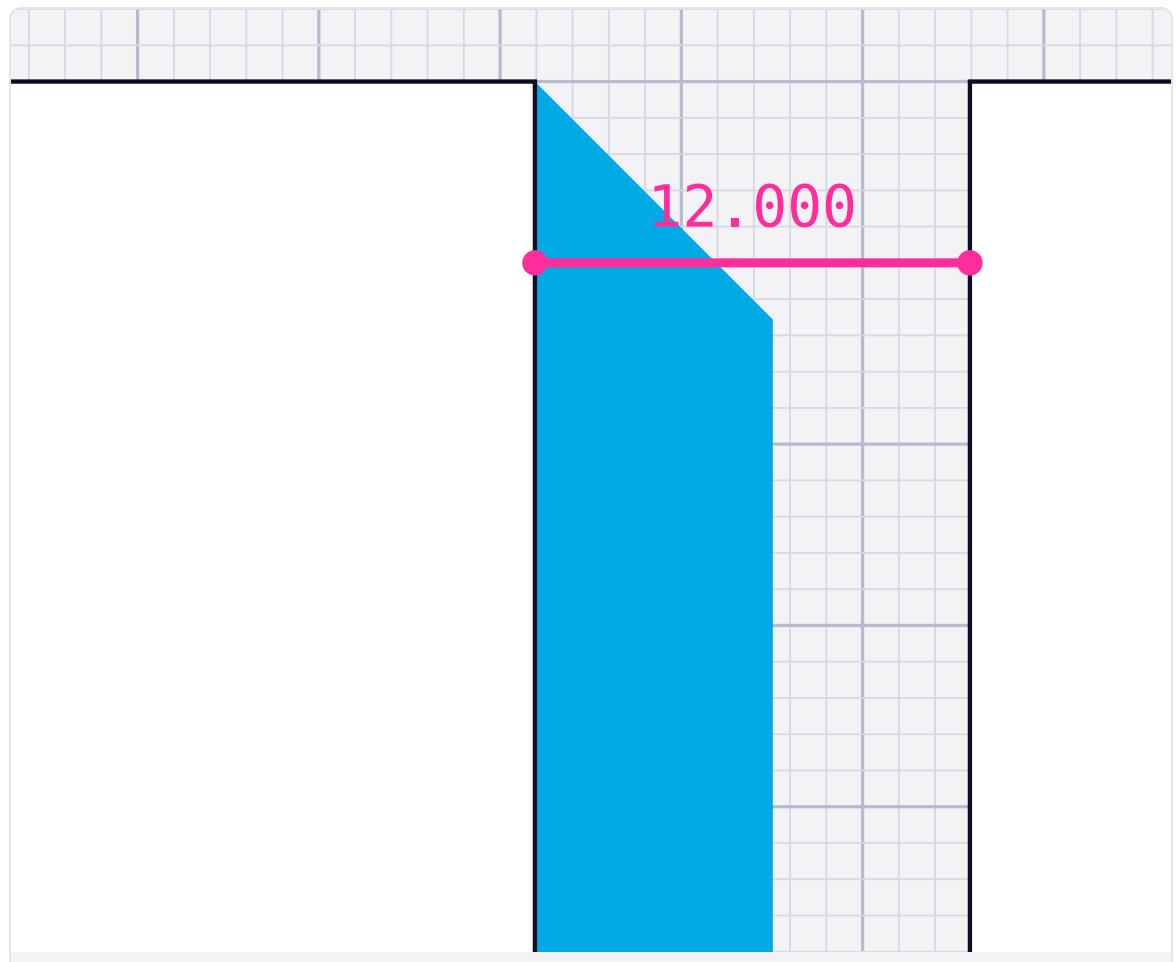
Letters are spaced face to face: 12 inside a word, 29 between words. Positions are exact.

Word	Letter	x left	x right	Advance
SAN	S	0	91.5	91.5
SAN	A	103.5	224.328125	120.828125
SAN	N	236.328125	353.78125	117.453125
JUAN	J	382.78125	479.5078125	96.7265625
JUAN	U	491.5078125	612.4140625	120.90625
JUAN	A2	624.4140625	745.2421875	120.828125
JUAN	N2	757.2421875	874.6953125	117.453125
BOWL	B	903.6953125	1009.421875	105.7265625
BOWL	O	1021.421875	1117.921875	96.5
BOWL	W	1129.921875	1321.25	191.328125
BOWL	L	1333.25	1440.4765625	107.2265625

7 · Clearances and minimum size



Tightest spot, A inner foot (1 square = 1 u). Separate shadow pieces 2.16 apart, shortest shadow edge 2.25, background channel 2.609.



U opening between the inner serifs: exactly 12, the same as the letter gap.

The smallest feature anywhere is 2.16 u. Keep it above what the process can hold to set a minimum cap height. These thresholds are typical starting points; confirm with the vendor.

Process	Smallest feature it holds	Minimum cap height
Offset / digital print, finest line ≈ 0.10 mm	0.10 mm	5.6 mm
Screen print / vinyl cut ≈ 0.30 mm	0.30 mm	16.7 mm
Embroidery satin ≈ 1.0 mm	1.0 mm	55.6 mm
Screen, keep smallest gap ≥ 1 px	1 px	56 px

Sizes at common cap heights

Cap height	1 u	Width with shadow	Height with shadow	Stroke 33	Serif	Shadow offset	Smallest feature
10 mm cap	0.0833 mm	120.59 mm	10.55 mm	2.750 mm	0.894 mm	0.547 mm	0.180 mm
25 mm cap	0.2083 mm	301.47 mm	26.37 mm	6.875 mm	2.235 mm	1.367 mm	0.450 mm
100 mm cap	0.8333 mm	1205.87 mm	105.47 mm	27.500 mm	8.939 mm	5.469 mm	1.800 mm
1 in cap	0.0083 in	12.06 in	1.05 in	0.275 in	0.089 in	0.055 in	0.018 in
1 ft cap	0.1000 in	144.70 in	12.66 in	3.300 in	1.073 in	0.656 in	0.216 in
8 ft cap (field)	0.8000 in	1157.63 in	101.25 in	26.400 in	8.581 in	5.250 in	1.728 in
60 px cap (screen)	0.5000 px	723.52 px	63.28 px	16.500 px	5.363 px	3.281 px	1.080 px

8 · Colour

Face · White
#FFFFFF · C0 M0 Y0 K0

Shadow · SJB Cyan
#00AAE7 · C86 M8 Y0 K0

Ground · SJB Navy
#0D0E21 · C86.92 M81.16 Y55.29 K74.73

The lettering is white with a cyan shadow and is always placed on navy. It has no background of its own, so on a light ground only the shadow would show.

9 · File and editing rules

- **File:** `SJB-Lettering-Exact.svg`. Groups SAN, JUAN and BOWL, then one group per letter holding `Shadow-X` under `Face-X`. No strokes, masks, clips or transforms. Straight lines only.

- **Scale** uniformly by cap height. Never stretch one axis, never redraw single letters, never recolour the shadow separately from the system.
- **Kern** by moving a whole letter group (shadow and face together). Keep the 12 / 29 face-to-face rhythm.
- **Exactness:** every coordinate is a multiple of 1/256 and is stored exactly in Illustrator and browsers. Avoid "simplify path" and snapping to pixel grid, which break it.
- **Changing a rule** (stroke, serif, shadow) means regenerating all letters from the construction, not editing points by hand. The shadow is always the full sweep of the face.

10 · Verification

Checked on the saved file with exact rational arithmetic and zero tolerance, then by four independent blind reviews. The last review found no defects.

Check	Result
Every face sits on cap 0 and baseline 120	PASS · 11 / 11
Edge directions	86 H · 72 V · 18 at 45° · 20 at 7:24 · 4 at 1:2 · nothing else
Horizontal edges on the 8 horizon lines	PASS · 0 off-line
Stems and strokes	33.000000 on every stem and 7:24 stroke (exact, width² = 1089)
Serifs	38 overhangs, all exactly 10.7265625
Corner cuts	all 17 × 17 at 45° (B waist excepted)
Symmetry	A, A2, U, W, O mirror-exact · S and O point-symmetric
Repeats	A2 = A and N2 = N, exact translations (face and shadow)
Shadow	equals the exact sweep of each face by (6.5625, 6.5625) · 0 mismatch
Coordinates	all finite decimals · faces on a 1/128 grid, shadows on 1/256 · nothing rounded
Topology	no repeated points, zero-length or collinear segments, self-intersections
Fill rule	outer contours one winding, counters the other · nonzero = evenodd (0 px differ)
Narrow features	shadow pieces ≥ 2.16 apart · background ≥ 2.609 · shadow segment ≥ 2.25
Review	own exact audit + 4 independent blind reviews · final review: 14 / 14 pass, no defects